

Experiment-7

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Course Code: -MLA0305

Slot-B

Title

Implementation of Temporal Difference Learning using TD(0)

Aim

To implement the **Temporal Difference (TD) Learning algorithm using TD(0)** for estimating the state-value function and to study how the value function is updated incrementally using the **TD error**.

Procedure

1. Define a Reinforcement Learning environment with a set of states and actions.
2. Define the transition function and reward for each state-action transition.
3. Initialize the state-value function to zero.
4. Select the learning rate and discount factor .
5. Start the agent from the initial state.
6. Select an action and observe the next state and reward.
7. Calculate the **TD target**:

$$R_t + 1 + \gamma V(S_t + 1)$$

8. Calculate the **TD error**:

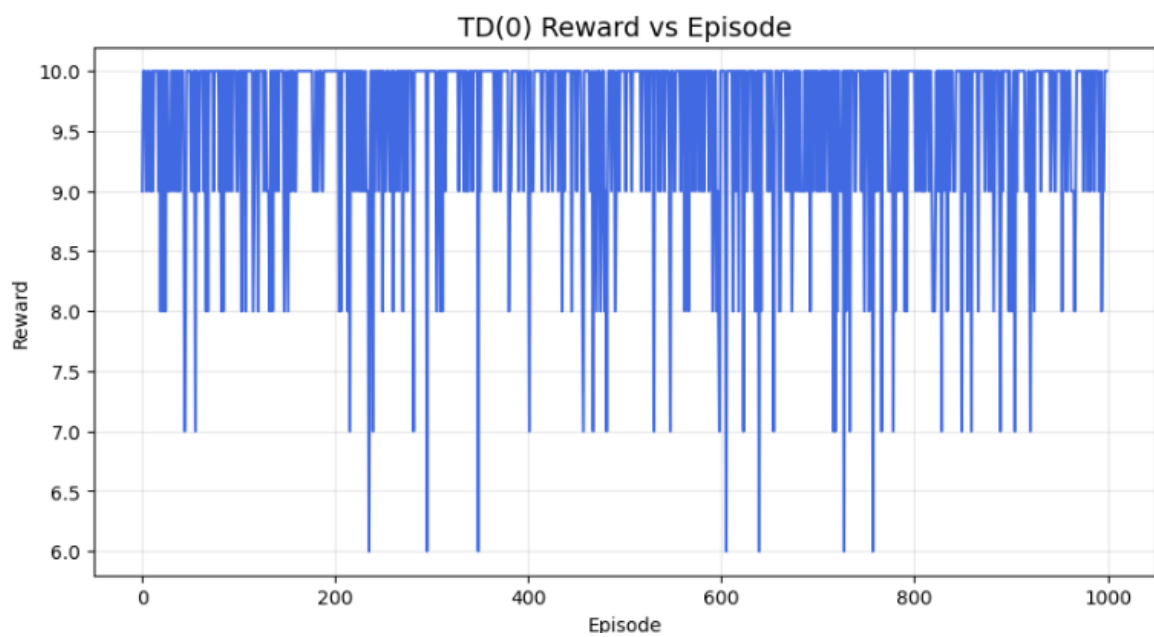
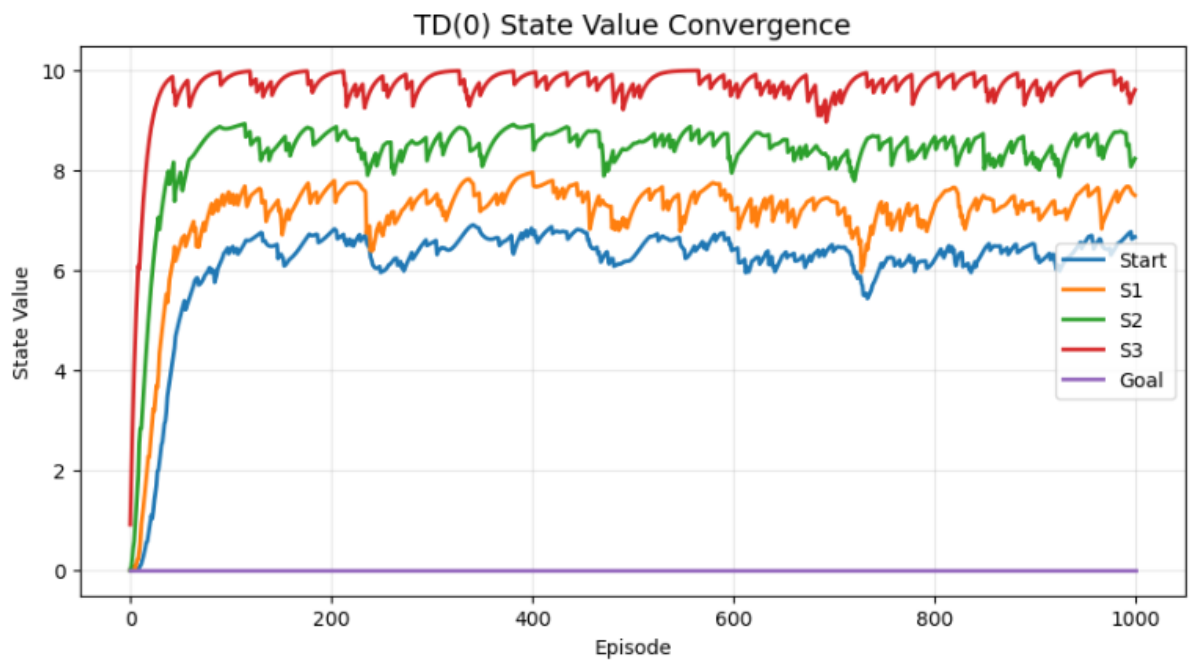
$$\delta t = R_t + 1 + \gamma V(S_t + 1) - V(S_t)$$

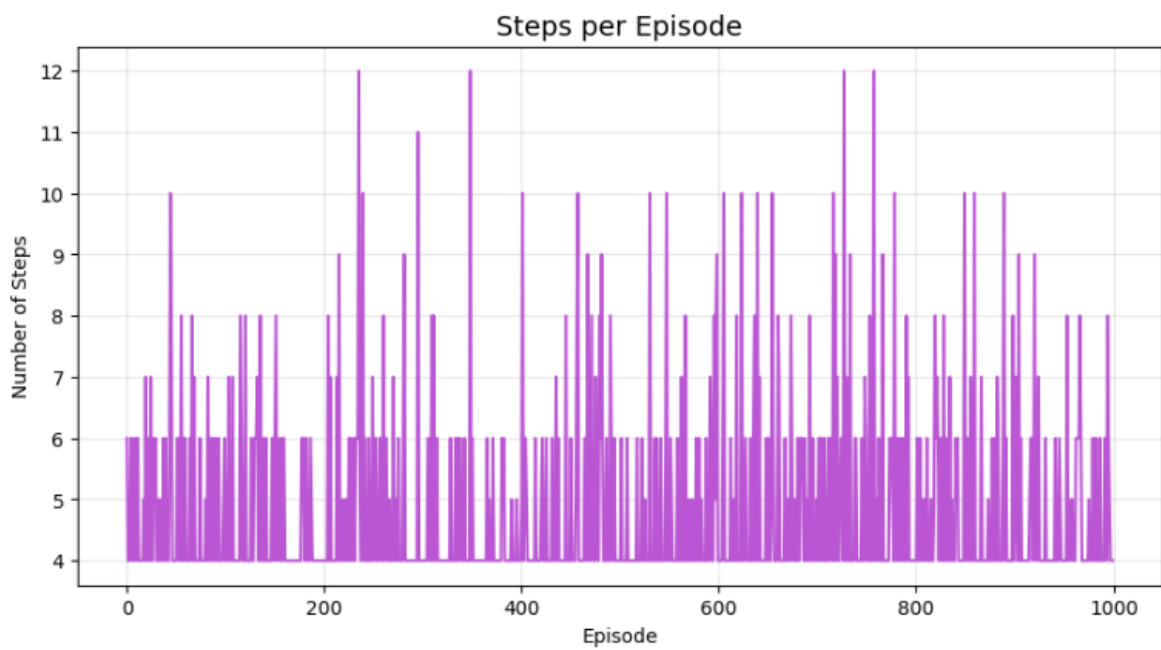
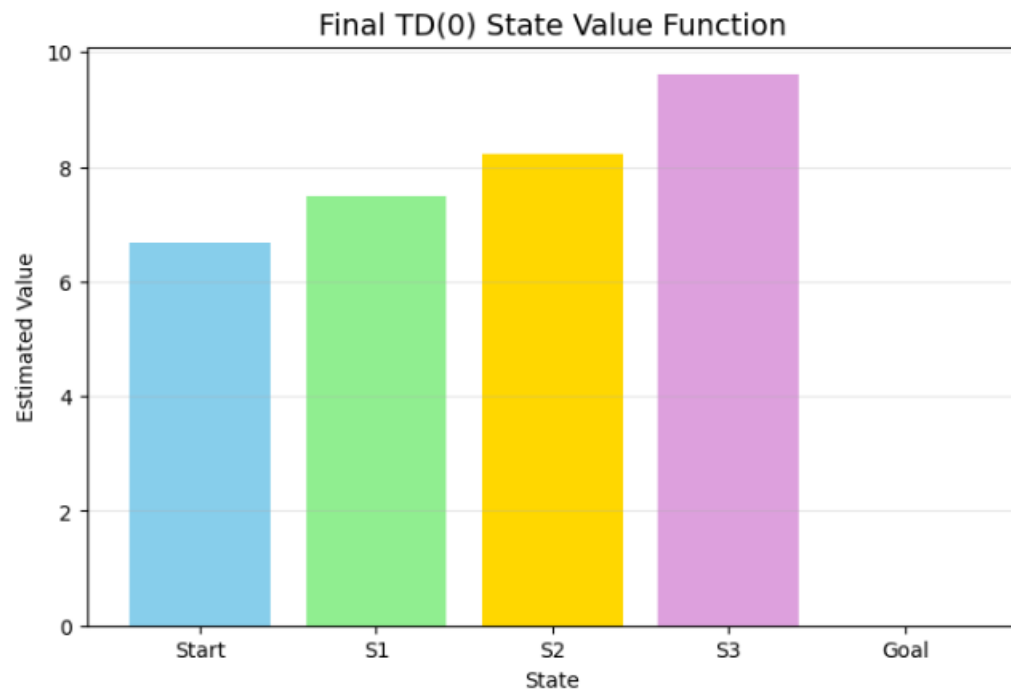
9. Update the current state's value using the TD(0) equation:

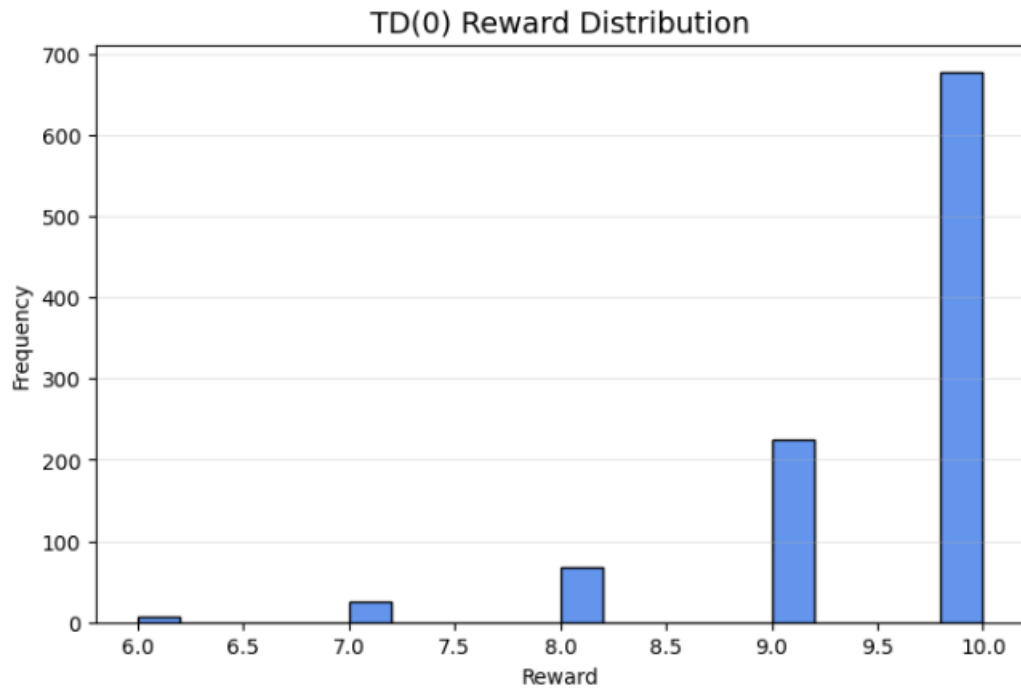
$$V(S_t) \leftarrow V(S_t) + \alpha [R_t + 1 + \gamma V(S_t + 1) - V(S_t)]$$

10. Move to the next state and repeat the process.
11. Continue the learning process for the specified number of episodes.
12. Record the reward, TD error, number of steps, and state values for every episode.
13. Analyze the convergence of the state-value function.
14. Calculate the average reward, TD error, success rate, and average episode length.
15. Display the results using summary tables and graphs.
16. Export the final state values and performance results as CSV files.

Visualizations







Summary Table

State	Initial Value	Final TD(0) Value	Rank
Start	0.00	Obtained from program	Obtained
S1	0.00	Obtained from program	Obtained
S2	0.00	Obtained from program	Obtained
S3	0.00	Obtained from program	Obtained
Goal	0.00	Obtained from program	Obtained

Result

Temporal Difference Learning using TD(0) was successfully implemented. The algorithm updated the state-value function after every state transition using the TD error. The value estimates gradually converged through repeated learning episodes. The experiment demonstrates that TD(0) can estimate state values incrementally without waiting for the complete episode to finish and without requiring a model of the environment.